

FeltSight

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Figure 1: Sensitizing the environment like a star-nosed mole with “FeltSight”. This bio-inspired haptic gloves extend tactile perception beyond the skin, enabling users to “feel” material textures at a distance. It enhances users’ tactile acuity while deemphasizing vision.

ABSTRACT

FeltSight is a mixed reality haptic experience that reimagines human perception by drawing inspiration from the tactile navigation of the star-nosed mole. Moving beyond traditional, vision-dominated interaction paradigms, FeltSight enables users to engage in meditative wandering guided by extended-range haptics with subtle visual cues. The system comprises a wearable haptic glove paired with an extended reality interface. As users reach toward objects in their environment, the glove’s vibration actuators—driven by audio-responsive patterns—simulate material textures, producing a sensation akin to touch. Meanwhile, the mixed reality interface offers a deliberately “reduced reality”, presenting nearby objects as dynamic point clouds that materialize only in response to exploratory hand gestures. By shifting perceptual focus from the visual to the tactile, FeltSight challenges ocularcentric sensory hierarchies and foregrounds an embodied, relational, and more-than-human mode of sensing.

Index Terms: Mixed Reality, Augmented Humans, Star-Nosed Mole, Haptic, Somaesthetics, Meditative Wandering, More-than-human Design

PROJECT URL

<https://reality.design/project/feltsight>

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1 DESCRIPTION OF ARTWORK

1.1 Thematic Statement

Vision dominates the hierarchy of the human sensory system, deeply embedded in our culture and known as *ocularcentrism*, which privileges visual perception as the primary pathway to knowledge. Donna Haraway’s concept of *tentacular thinking* critiques this anthropocentric, vision-dominated worldview, advocating instead for forms of knowing that emerge through touching, feeling, and probing [1]. When we zoom out to look at more-than-human perception, we can see other species construct rich perceptual environments through different sensory hierarchies—some prioritize touch. For example, the star-nosed mole is functionally blind yet “sees” through touch—its unusual star-shaped nose contains over 100,000 nerve endings, forming an extremely sensitive tactile organ that constructs its mental map via touch, shaping the mole’s *umwelt* [2].

1.2 Artwork

We introduce *FeltSight*, a mixed reality system that extends tactile sensation by equipping users with haptic sensing and shifting perceptual hierarchy from vision to touch. This system features a custom haptic glove paired with Apple Vision Pro, allowing users to “feel” their surroundings at a distance. Drawing from *Embodying the Monster* [7], the gloves’ design mimics the distinctive nose structure of the star-nosed mole. These gloves contain vibration actuators in each finger that enable sensing the texture of physical objects from a distance. This effectively extends touch beyond the skin—as a finger approaches or moves along an object at a distance without physical contact, the glove generates material-specific vibrations that simulate touching that surface [5]. Haptic feedback dynamically adapts to finger movement. Visual information in the headset is deliberately minimized, displaying objects only as LiDAR-scanned point clouds within a small detection sphere around the hand. By weakening visual perception while enhancing tactile sensation, FeltSight encourages users to engage in



Figure 2: Participants often walk with their backs bent, relying on information from the ground.

active, mindful tactile exploration rather than passive observation, fostering deeper engagement with the surrounding space.

1.3 User Experience and Embodied Response

The user experience of FeltSight resembles meditative wandering inspired by ancient walking-meditation practices [8] that help users regain awareness of materials and surroundings. By intentionally minimizing visual input, FeltSight guides the audience to focus on subtle tactile sensations, encouraging them to slow down and pay attention to their surroundings. The glove gently redirects the user's attention toward fine simulated texture details through haptic feedback—the branches of a nearby tree, the surface of a rock, the fabric of someone's clothing—that might otherwise be overlooked. Users naturally decrease their pace to perceive the surrounding materials better. In this context, vision becomes a secondary sense, used sparingly and in service of tactile inquiry.

As a more-than-human design [6, 9], FeltSight momentarily immerses audiences in a tactile *umwelt*, encouraging heightened sensitivity to their surroundings. Users become aware of *umgebung*—the world that exists but typically goes unperceived [2]. This inversion of the traditional sensory hierarchy transforms meditative wandering into a reflection on alternative ways of perceiving the world. Rooted in technology-mediated digital touch [4] and somaesthetic design [3], FeltSight poetically expands the human sensorium through bio-inspired design. It invites participants into a deeply embodied mode of meditative wandering—one where the boundaries between vision and touch, human and animal, self and technology become productively blurred.

We invited participants experience FeltSight. Deprived of reliable sight, we found that participants naturally slow their pace to a cautious, navigation deviated from linear paths; it became aimless, nonlinear, and organic—guided by shifting haptic sensations that invited detours toward unseen textures. These altered body mechanics suggest a redistribution of sensory priority: as vision receded, touch took precedence, many bent or crouched, adopting a burrowing, animal-like posture. By enabling relational, tactile-first encounters, the system fostered more-than-human sensibilities, inviting participants into deeper entanglements with their environment.

1.4 Relevance to the Theme

FeltSight stages a reimagined “alternative *umwelten*” present in which remote touch organizes attention. By granting audiences a temporary tactile *umwelt*, it bridges current MR capabilities with a near-future practice of multisensory, care-oriented navigation. This reprioritization of the sensory hierarchy resonates with the IEEE VR XR Gallery's theme: Bridging the Reimagined Present through Immersive Experiences.

2 ARTIST INFORMATION

Danlin Huang is an undergraduate student at the China Academy of Art, majoring in Industrial Design. Her media artworks include “Body Oracle” and “Foldiverse.” As an artist, speculative design

researcher, and AR developer, her academic papers have been published in top conferences such as SIGGRAPH Asia and CHI PLAY. Her works have been exhibited at the China Digital Art Exhibition and the China National Art Exhibition, among others. Huang's work has earned her multiple accolades, including the Platinum Creative Award, the Gold Award at the Rokid University AR Innovation Challenge, the NCDA Gold Award, and the CADA Gold Award.

Botao ‘Amber’ Hu is a social computing researcher and experiential futures designer. He is currently a PhD candidate in Human Centred Computing group at University of Oxford's Department of Computer Science. He directs Reality Design Lab (<https://reality.design>), an independent interdisciplinary research and design lab. His primary focus is designing socially experiential futures using mixed reality as the main medium. He also serves as a visiting lecturer at the China Academy of Art.

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